

Does no shield mean no libraries?

Yes. But not necessarily. There are enthusiastic developers who've written their own libraries for things such as sensor use. A quick search online can reveal many such libraries.

There are many tutorials available online to help make a shield, if you're keen to develop your own shield and make it available to other people. All you need is some knowledge of PCB design.

Shields available

Many different types of shield are available for various purposes. Given below is a list of available shields and the purpose they're used for.



Arduino™ Motor Shield

Arduino™ Motor Shield

Motors can't be directly connected to the microcontroller as they draw a lot of power. No microcontroller is capable of providing power to a DC motor. However, what can control a motor are circuits like H-bridge built using MOSFETs (metal oxide semiconductor field effect transistors).

The 'Motor Shield' is based on an IC L298, commonly referred to as 'motor driver'. It can supply sufficient power to drive motors with operating voltage of 5-12 volts.

The shield is sufficient to drive a low-power DC motor with up to 2A current and has current sensing feature at 1.65A. For higher power motors, you'll need custom design hardware. For such a custom design, you can search online for motor drivers.

The motor shield can drive solenoids, relays, DC motors and stepper motors.

Arduino™ Ethernet Shield

To build the Internet of Things, your device will need networking capabilities.

This is where the 'Ethernet Shield' comes in. It adds networking



Arduino™ Ethernet Shield